

CS 442: Algorithm Design and Analysis

Fall 2026



Course Information

Credits: 5

Time & Location: Hybrid.

- Mo 12:30PM - 2:20PM @ Snohomish 119
- We 12:30PM - 2:20PM @ Snohomish 119
- Fr 12:30PM - 1:20PM Online over zoom

Instructor Information

Instructor: Stephen Jonany

Email: stephen.jonany@edmonds.edu

Office Hours: By appointment

Best way to contact: By email. I will respond within 24 hours, except during holidays

Texts, Materials, and Supplies

Required textbook. None. Supplemental materials will be provided in the course page.

Required materials

- Access to a computer with Python 3.9 or higher and an editor/IDE
- Access to the most recent Google Chrome browser.
- A working microphone and webcam

Canvas: Edmonds College uses [Canvas](#) for online learning. To use Canvas, you'll need:

- Access to a computer or internet-enabled device
- A reliable internet connection

- A web browser: Chrome works best

Technical Support for students is available at itsupport.edmonds.edu.

About this Course

Purpose. This course equips students with the foundational skills to design, analyze, and prove the correctness of efficient computational algorithms across multiple design paradigms.

Enrollment Requirements (Prerequisites): Completion of CS 302 and CS 334 with a grade of 2.5 or higher or instructor permission.

Course description: Principles of design of efficient algorithms. Topics include: recursion, divide and conquer, dynamic programming, greedy method, data structure selection, correctness and analysis of algorithms.

Course modality: Hybrid course. A hybrid course is a combination of traditional classroom activities and online instruction. We will meet in person on Mondays and Wednesdays. We will meet online on Zoom on Fridays.

Learning Objectives

Upon successful completion of this course, students will be able to:

- Model word problems as computational problems.
- Determine an appropriate algorithm design paradigm for a new problem.
- Design an algorithm using a variety of algorithm-design paradigms (including greedy algorithms, divide and conquer, dynamic programming, flow modeling, and others).
- Prove your algorithm produces the correct answer.
- Reduce between a new problem and a known problem (for showing hardness or for reusing existing algorithms)
- Identify and cope with computational problems that are infeasible.
- Consider the implications of modeling decisions in the real world.

Grading Information (How Your Learning Will Be Assessed)

Grade distribution

- Weekly homework (20%)
- Weekly in-person mastery quizzes (50%)
- Mock interview (10%)
- Midterm (10%)
- Finals (10%)

In-person. Most of the grades are on in-person activities. Only **20%** is delegated to homework. So, please make sure to attend **every class**.

Drops. Your lowest quiz score will be dropped. Your lowest homework score will also be dropped.

Your final grade is calculated based on the number of points you have earned. To support everyone's success, course grades may be adjusted upward based on overall class performance. They will never be adjusted downward.

GPA	Percentage
4.0	95%-100%
3.7 - 3.9	90%-95%
3.3 - 3.6	85%-89%
3.0 - 3.2	80%-84%
2.7 - 2.9	75%-79%
2.3 - 2.6	70%-74%
2.0 - 2.2	65%-69%
1.7-1.9	60%-64%
1.1-1.6	55%-59%

Assignment Information

Mastery quizzes

Content. These will test concepts tackled in the homework. A diligent student who has done the homework should be able to ace these quizzes.

Mastery, retakes. My goal is to encourage mastery over materials. You will only get credit if you get **all** the questions right, to show that you have really mastered the concept. Each quiz can be attempted up to **3 times**, but they will follow the rigid schedule whereby it's administered on Monday, Wednesday, and the following Monday. E.g. In Week 1, we will learn about divide and conquer algorithms. You will get to attempt the quizzes on Week 2 Monday, Week 2 Wednesday, and Week 3 Monday.

Proctoring. These will be done in person on Mondays and Wednesdays during lecture time, without computer, and should take around 15 minutes. I allocate 25 minutes for quizzes so you can retake a previous quiz.

Drops. Your lowest quiz score will be dropped.

Homework

Content. There will be a weekly homework due on Sundays. There might be some conceptual questions, but the majority of the work would be to solve a set of curated **Leetcode** problems. By the end of this course, the hope is that you will have a diverse enough repository of problems to tackle algorithm interviews. Note that your Monday quiz covers the homework due the day before on Sunday.

Grading. Each section is worth 1 point: 0 if not submitted, 1 if attempted. I will leave feedback for each homework personally, but correctness-based grading will be deferred to the mastery quizzes.

Self-driven learning. You will also be able to **self-grade** by chatting against a [Gemini Gem](#) (Google's AI chatbot) that I have prepared for each homework beforehand. This bot will already contain the answer key, but is curated in a way that fosters learning on your end.

Late policy. Each student receives **3 late days** for use on homework. A late day allows you to submit an assignment up to 24 hours late without penalty. For example, you could use a late day and submit a homework due Sunday at 11 pm on Monday by 11 pm with no penalty. You can only use **one late day** per assignment. Once a student has used up all the late days, any late assignment will be given **0 credit**. Students will not be given extensions unless they have extenuating circumstances as decided by the instructor.

- **But really, don't be late.** Note that each homework is graded **leniently**, and the learnings will feed into the mastery quizzes. As such, you should **not** aim to use a late day. Submitting a decently-attempted homework can still net you a full credit.

Drops. Your lowest homework score will be dropped.

Mock interview

Every Friday we're going to have a mock interview based on this week's lecture. I will be the interviewer, and 1-2 students will be the interviewee. You get a course's worth of **10% full participation credit** simply for being the interviewee for that week.

Exams: Midterm, Final

All exams will be administered individually in person, on paper. No computers are allowed. Think of it as a real job interview.

Make-up exams will not be given except in case of a serious emergency. If you must miss an exam, even if you are sick or injured, you must contact Stephen before the exam (or arrange for someone to do so). You must show evidence that you are physically unable to take the exam, such as a clear and specific doctor's note mentioning the date, exam, and reason. No make-ups

will be granted for personal reasons such as travel, personal hardship, leisure, or to ease exam week schedules.

Course Calendar

Week	Topic	Evaluations
1: September 21, 2026	Divide and conquer.	Quiz 1: Runtime Mock interview 1 on Fri HW 1 (Sunday): Divide and conquer
2: Sep 28	Dynamic programming. 1D	Quiz 2: Divide and conquer Mock interview 2 HW 2 (Sunday): DP 1D
3: Oct 5	Dynamic programming. 2D	Quiz 3: DP 1D * Oct 9 Fri, no class. HW 3 (Sunday): DP 2D
4: Oct 12	Streaming algorithm. Sliding window	Quiz 4: DP 2D Mock interview 3 HW 4: Streaming
5: Oct 19	Greedy algorithms. Optimality arguments	Quiz 5: Streaming Mock interview 4 HW 5: Greedy
6: Oct 26	Midterm recap	Midterm on Wednesday Oct 28. Quiz 6: Greedy *Oct 30 Fri, no class HW 6: Midterm recap
7: Nov 2	Graph traversals	No quiz Mock interview 5 HW 7: Graph traversals
8: Nov 9	Dynamic programming on graphs	Quiz 7: Graph traversals * Nov 11 Wed no class No mock interview, Fri is lecture. HW 8: DP on graphs I
9: Nov 16	Dynamic programming on graphs II, Greedy on graphs	Quiz 8: DP on graphs I Mock interview 6 HW 9: DP on graphs II
10: Nov 23	P vs NP intro	Quiz 9: DP on graphs II * Thanksgiving, so we only have 1

		class this week HW 10: P vs NP
11: Nov 30	P vs NP. Final review.	Quiz 10: P vs NP No mock interview, Fri is lecture No HW. Good luck on finals!
12 (Finals week)	N/A	Finals

Holidays. See the college calendar [here](#).

- Oct 9. Non-instructional day
- Oct 30. Non-instructional day
- Nov 11. Veteran's day.
- Nov 25-28. Thanksgiving

Projected weekly cadence. Aside from the midterms and finals, your weekly assignment schedule will look something like this:

- **Sunday.**
 - **Homework is due.** Topic from this week. Will be quizzed tomorrow.
- **Monday.**
 - **Quiz.** New mastery quiz in class, which is based on the submitted homework. Optionally retake the mastery quiz from last week.
 - **New hw.** Start doing the homework due on Sunday
- **Wednesday.**
 - **Quiz retakes.** Just retakes.
- **Friday.** Online mock interview.

Final Exam and Last Meeting of Class

TBD. Once finalized at the [college-level calendar](#), I will announce.

Important Dates

First day of classes: Sep 21

Last day to withdraw: Nov 10

Midterm: Oct 28 Wed.

Final: TBD. Once finalized at the [college-level calendar](#), I will announce.

Other important dates: [Edmonds College Academic Calendar](#)

Classroom Community Expectations

Expectations for Instructor

You can expect that I will do everything I can to help you learn. I will provide timely, meaningful feedback on your work and answer emailed questions within 24 hours except over holidays. I will be available by appointment to work individually with you over Zoom or Meet.

Expectations for Students (How to Succeed in this Course)

Complete all the assigned work, and follow the Policy on Use of Generative AI. If you are unsure, please **ask**, in class, right after class, or asynchronously through email.

Attendance, Participation, and Classroom Climate

In-person attendance. We have a graded assignment (quiz, mock interview) **every** class time. So, you are expected to attend all the course sessions.

Withdrawal. If you wish to withdraw from this course it is your responsibility to do so within the allotted time frame. If you stop participating in the class, it will not result in an Instructor Withdrawal (V); you will receive the appropriate grade based on the number of points earned for the quarter.

Diversity and Inclusion Statement

It is my intent that students from all backgrounds and perspectives be well served by this course, that students' learning needs be addressed both in and out of class, and that the diversity that students bring to this class be viewed as a resource, strength and benefit. It is my intent to present materials and activities that are respectful of diversity: gender, sexuality, disability, age, socioeconomic status, ethnicity, race, religion, and culture. Your suggestions are encouraged and appreciated. Please let me know ways to improve the effectiveness of the course for you personally or for other students or student groups.

Statement on Academic Integrity

Edmonds College students shall demonstrate Academic Integrity. Instructors are expected to report all violations of Academic Integrity (cheating and plagiarism) to the College. The College's database of such incidents will be monitored by the Office of the Vice President for Student Services. Data will be maintained for three years. Evidence of repeat incidents will result in additional action by the Office of the Vice President for Student Services as governed by the Student Code of Conduct.

Policy on Academic Integrity

In-person assessments. All in-person assessments (mastery quizzes, midterms, finals) must be done individually, without the usage of any computing devices or any notes.

Homework. You may collaborate and use external resources, but (1) you are encouraged to do the homework **alone first** and (2) you must **attribute** these external resources when you do use them. The homework guideline will be very explicit on how long you should work on your own first v.s. resorting to external help. You don't have to cheat. I give 100% credit on attempts, even if they are incorrect. Cheating on homework will also only hurt you. If you cheat, I won't be able to give you the proper feedback to help you learn, nor adjust the difficulty of the quizzes. You will also likely fail on the in-person mastery quizzes, which test concepts tackled in the homework.

Grade implication. If you cheat on a mastery quiz or exams, you will get a grade penalty ranging from half credit on the assignment to 1.0 lower in the course (or a failing grade in the course if the incident is on the midterm or final). A second incident in this class will result in a grade penalty ranging from a zero on the assignment to a failing grade in the course.

Policy on Use of Generative AI

Motivation. Remember that you will get full credit on homework even on incorrect answers. You do **not** have to cheat. On the other hand, remember that **80%** of the course grade (quizzes, exams, mock interviews) will test your ability to answer without AI. So, use Generative AI to help you only **after** you have done the hard work of making your first attempt without AI. **Why:** (1) [Unregulated usage of AI harms learning](#) (2) Even in the age of AI, your [AI-independent ability matters](#).

When: Self-study. You are highly encouraged to use Generative AI to help you (1) understand the lectures and (2) get you unstuck on homework.

Which AI? You should only use either (1) The generic [guided learning](#) mode on Gemini, which comes with your school account or (2) the homework-specific [Gemini Gem](#) that I will have pre-created for you.

Usage. Please cite AI use. I will not police you, but this is so I can give you better feedback and tune quiz difficulty. I recommend these two Gemini tools because they're built to teach rather than hand you answers. Generic chatbots will hurt you on the quizzes.

Services for Students with Disabilities

[Services for Students with Disabilities](#) (SSD), located in MLT 159, ensures that programs at Edmonds College are accessible and usable by students regardless of ability. If you require an accommodation for a disability, or if you have struggled with

learning in the past and might benefit from a consultation, please contact SSD online or by phone: 425.640.1320 or email: ssdmail@edmonds.edu.

Resources for Students

Edmonds College offers a number of student resources to help you succeed. Asking for help, whether you are struggling or not, is a normal part of being a student. You may find it helpful to browse this [Student Resources Guide](#).

Resources for Students by Category

Academic Support

[Learning Support Center](#): free tutoring for math, writing, and other classes

[TRIO Student Support Services](#): tutoring and other support for students who are the first in their family to go to college

[e-Tutoring](#): free online tutoring for math and writing

[Library](#): 24/7 live chat help for research and citations, free online access to ebooks, articles, and databases

[Campus Study Spaces](#): A map of areas and rooms where students can study and attend online synchronous class sessions.

[STEM Study Room](#): free drop-in tutoring for Math, Biology, Chemistry, Physics, Computer Science, and Engineering

Basic Needs Support (food, housing)

[Edmonds College Food Pantry](#): on-campus resource for students and employees facing food insecurity

[Emergency Funding and Resources](#): help with housing costs or other financial difficulties

Health and Well-Being

[Center for Student Cultural Diversity and Inclusion](#): meet with a student success coach

[Counseling and Resource Center](#): personal counseling and resources

[Violence Prevention](#): confidential support for those who have experienced sexual assault, dating violence, domestic violence, or stalking

[Wellness Center](#): a quiet space to study, nap, or relax

Technology Assistance

[Edmonds College IT Help Center](#): help with Canvas, CTC Link, Google apps, and other technologies

[Student Technology Advice and Resource Team \(START\)](#): a student-led resource focused on helping students understand college-provided technology and learn how to put that technology to its best use.

[Canvas Log-in and Orientation](#): how to use Canvas and reset your password

[Student Printing Information](#): where to print on campus

[Computer Labs on Campus](#): location and use of on-campus computer labs for use by students

Veterans Resources

[The Veterans Resource Center](#): services for Veterans including advising, help with use of VA training or benefits

Resources for Non-immigrant Visa Students

[Office of International Programs](#): advising and activities for non-immigrant visa students

Student Government and Campus Involvement

[Center for Student Engagement and Leadership](#): campus leadership opportunities

College Policies and Guidelines

- [Non-Discrimination Statement](#): Edmonds College does not discriminate on the basis of race; color; religion; national origin; sex; disability; sexual orientation; age; citizenship, marital, or veteran status; or genetic information in its programs and activities.
- [Institutional and Instructor Withdrawal Policies](#). Withdrawing from class is the process of formally dropping a class or classes after the quarter has started.

The college may withdraw a student from a class or classes if the student has been suspended for poor conduct or a lack of academic achievement, or if a

student has not paid tuition. Tuition refunds may not be provided for institutional withdrawals.

- **Emergency Preparedness:** The [Triton Alert System](#) will be used to send notifications regarding campus closures, emergency situations, or evacuation orders in the event of an emergency or inclement weather. Edmonds College students and employees are automatically enrolled to receive Triton Alerts through their college email addresses. Sign up to receive text and voice messages on your mobile or home phones and/or additional email notifications to personal email addresses. I will send a message through canvas and email to discuss the course-specific changes.
- Please see [Student Rights, Freedoms, and Responsibilities](#) for guidelines for student conduct and information on disciplinary procedures.
- **Reporting Gender Discrimination and Sexual Assault:** I hope that your experience in my class and at Edmonds College is fair and respectful of your individuality, including your gender identity and sexuality. If you do experience unfair treatment, discrimination, or harassment based upon your gender identity, or if you are the victim of sexual misconduct, please [report the incident](#) so that you can receive help.

As your instructor, I am required to report any incidents of gender discrimination or sexual harassment to the Edmonds College's Title IX Officer. If you do not wish to report an incident but would like to speak with a confidential counselor, please contact the [Counseling and Resource Center](#).

For more information about or examples of gender discrimination, please see [Title IX and Gender Discrimination](#).

- **Reporting Discrimination or Inappropriate Behavior:** If you experience discrimination or inappropriate behavior, please complete the [Bias Incident Form](#) or contact your instructor or any college employee. If this is an emergency, please call 911 and Campus Security at 425-754-0154.
- **Student Conduct Guidelines and Procedures:** Please see [Student Rights, Freedoms, and Responsibilities](#) for guidelines about student conduct and information on disciplinary procedures.
- **Accommodations for Religion/Conscience:** Students who will be absent from course activities due to reasons of faith or conscience may seek reasonable accommodations so that grades are not impacted. Such requests must be made

within the first two weeks of the quarter and should follow the procedures listed in the college's [Absence for Reasons of Faith or Conscience policy \(SS 8.01Pr\)](#)

One Last Note

This syllabus is intended to give students guidance in what may be covered during the term and will be followed as closely as possible. However, the instructor may choose to modify, supplement, and make changes to the course in the event of extenuating circumstances, by mutual agreement, and/or to ensure better learning.